

the subgraph induced by the edge set $E(\mathcal{T})$ is a triangle-forest-structure, then

$$rc(L(G)) \leq n_2 - t.$$

Moreover, the bound is sharp.

Theorem 2.36 [43] *If G is a connected graph, $\mathcal{T}$ is a set of t edge-disjoint triangles that cover all but n'_2 inner vertices of G and c is the number of components of the subgraph $G[E(\mathcal{T})]$, then*

$$rc(L(G)) \leq t + n'_2 + c.$$

Moreover, the bound is sharp.

Theorem 2.37 [43] *Let G be a connected graph with m edges and m_1 pendant 2-length paths. Then*

$$rc(L^2(G)) \leq m - m_1.$$

The equality holds if and only if G is a path of length at least 3.

In the proofs of the above three theorems, Li and Sun used the particular structure of line graphs and the observation: If G is a connected graph and $\{E_i\}_{i \in [t]}$ is a partition of the edge set of G into connected subgraphs $G_i = G[E_i]$, then $rc(G) \leq \sum_{i=1}^t rc(G_i)$ (see [42]).

The above three theorems give upper bounds for rainbow connection number of $L^k(G)$ ($k = 1, 2$) according to some parameters of G . One may consider the relation between $rc(G)$ and $rc(L(G))$.

Problem 2.38 *Determine the relationship between $rc(G)$ and $rc(L(G))$, is there an upper bound for one of these parameters in terms of the other?*

One also can consider the rainbow connection number of the general iterated line graph $L^k(G)$ when k is sufficiently large.

Problem 2.39 *Consider the value of $rc(L^k(G))$ as $k \rightarrow \infty$, is it bounded by a constant? or, does it convergence to a function of some graph parameters, such as the order n of G ? ■*

For Problem 2.39, we know if G is a cycle C_n ($n \geq 4$), then $L^k(G) = G$, so $rc(L^k(G)) = \lceil \frac{n}{2} \rceil$. But for many graphs, we know, as k grows, $L^k(G)$ will become more dense, and $rc(L^k(G))$ may decrease.

An *intersection graph* of a family of sets $\mathcal{F}$, is a graph whose vertices can be mapped to the sets in $\mathcal{F}$ such that there is an edge between two vertices in the graph if and only if the corresponding two sets in $\mathcal{F}$ have a non-empty intersection. An *interval graph* is an intersection graph of intervals on the real line. A *unit interval graph* is an intersection

graph of unit length intervals on the real line. A *circular arc graph* is an intersection graph of arcs on a circle. An independent triple of vertices x, y, z in a graph G is an *asteroidal triple* (AT), if between every pair of vertices in the triple, there is a path that does not contain any neighbour of the third. A graph without asteroidal triples is called an *AT-free graph* [20]. A graph G is a *threshold graph*, if there exists a weight function $w : V(G) \rightarrow \mathbb{R}$ and a real constant t such that two vertices $u, v \in V(G)$ are adjacent if and only if $w(u) + w(v) \geq t$. A bipartite graph $G(A, B)$ is called a *chain graph* if the vertices of A can be ordered as $A = (a_1, a_2, \dots, a_k)$ such that $N(a_1) \subseteq N(a_2) \subseteq \dots \subseteq N(a_k)$ [56]. In [11], Chandran, Das, Rajendraprasad and Varma investigated the rainbow connection numbers of these special graph classes. They first showed a result concerning the connected two-way dominating sets.

Theorem 2.40 [11] *If D is a connected two-way dominating set in a graph G , then*

$$rc(G) \leq rc(G[D]) + 3.$$

They also proved that every connected graph G of order n and minimum degree δ has a connected two-step dominating set D of size at most $\frac{3(n-|N^2(D)|)}{\delta+1} - 2$.

From Theorem 2.40, the following result can be derived.

Theorem 2.41 [11] *Let G be a connected graph with $\delta(G) \geq 2$. Then*

- (i) *if G is an interval graph, $\text{diam}(G) \leq rc(G) \leq \text{diam}(G) + 1$, in particular, if G is a unit interval graph, then $rc(G) = \text{diam}(G)$;*
- (ii) *if G is AT-free, $\text{diam}(G) \leq rc(G) \leq \text{diam}(G) + 3$;*
- (iii) *if G is a threshold graph, $\text{diam}(G) \leq rc(G) \leq 3$;*
- (iv) *if G is a chain graph, $\text{diam}(G) \leq rc(G) \leq 4$;*
- (v) *if G is a circular arc graph, $\text{diam}(G) \leq rc(G) \leq \text{diam}(G) + 4$.*

Moreover, there exist threshold graphs and chain graphs with minimum degree at least 2 and rainbow connection number equal to the corresponding upper bound above. There exists an AT-free graph G with minimum degree at least 2 and $rc(G) = \text{diam}(G) + 2$, which is 1 less than the upper bound above.

Recall that the concept of rainbow connection number is of great use in transferring information of high security in multicomputer networks. Cayley graphs are very good models that have been used in communication networks. So, it is of significance to study the rainbow connection numbers of Cayley graphs. Li, Li and Liu [38] investigated the rainbow connection numbers of cayley graphs on Abelian groups.

Let Γ be a group, and let $a \in \Gamma$ be an element. We use $\langle a \rangle$ to denote the cyclic subgroup of Γ generated by a . The number of elements of $\langle a \rangle$ is called the *order* of a , denoted by $|a|$. A pair of elements a and b in a group commutes if $ab = ba$. A group is *Abelian* if every pair of its elements commutes. A *Cayley graph* of Γ with respect to S is the graph $C(\Gamma, S)$ with vertex set Γ in which two vertices x and y are adjacent if and only if $xy^{-1} \in S$ (or equivalently, $yx^{-1} \in S$), where $S \subseteq \Gamma \setminus \{1\}$ is closed under taking inverse [52].

Theorem 2.42 [38] *Given an Abelian group Γ and an inverse closed set $S \subseteq \Gamma \setminus \{1\}$, we have the following results:*

- (i) $rc(C(\Gamma, S)) \leq \min\{\sum_{a \in S^*} \lceil |a|/2 \rceil \mid S^* \subseteq S \text{ is a minimal generating set of } \Gamma\}$.
- (ii) *If S is an inverse closed minimal generating set of Γ , then*

$$\sum_{a \in S^*} \lceil |a|/2 \rceil \leq rc(C(\Gamma, S)) \leq src(C(\Gamma, S)) \leq \sum_{a \in S^*} \lceil |a|/2 \rceil,$$

where $S^* \subseteq S$ is a minimal generating set of Γ .

Moreover, if every element $a \in S$ has an even order, then

$$rc(C(\Gamma, S)) = src(C(\Gamma, S)) = \sum_{a \in S^*} |a|/2.$$

They also investigated the rainbow connection numbers of recursive circulants (see [50] for an introduction to recursive circulants).

Let G be an r -regular graph with n vertices. G is said to be *strongly regular* and denoted by $SRG(v, k, \lambda, \mu)$ if there are also integers λ and μ such that every two adjacent vertices have λ common neighbours and every two nonadjacent vertices have μ common neighbours. Clearly, a strongly regular graph with parameters (v, k, λ, μ) is connected if and only if $\mu \geq 1$. In [1], Ahadi and Dehghan derived the following result: For every connected strongly regular graph G , $rc(G) \leq 600$. As each strongly regular graph is a graph with $diam(G) = 2$, from our next subsection (Theorem 2.51), we know $rc(G) \leq 5$ if G is a strongly regular graph with parameters (v, k, λ, μ) , other than a star [39]. But 5 may not be the optimal upper bound, so one may consider the following question.

Question 2.43 [1] *Determine $\max\{rc(G) \mid G \text{ is an } SRG\}$.*

There are other results on some special graph classes. In [14], Chartrand, Johns, McKeon and Zhang investigated the rainbow connection numbers of cages, and in [33], Johns, Okamoto and Zhang investigated the rainbow connection numbers of small cubic graphs. The details are omitted.

2.4 For dense and sparse graphs

For any given graph G , we know $1 \leq rc(G) \leq src(G) \leq m$. Here a graph G is called a *dense graph* if its (strong) rainbow connection number is small, especially it is close to 1; while G is called a *sparse graph* if its (strong) rainbow connection number is large, especially it is close to m . By Proposition 2.1, the cases that $rc(G) = 1, src(G) = 1$ and $rc(G) = m, src(G) = m$ are clear. So we want to investigate other cases.